

LLM Dunyasin

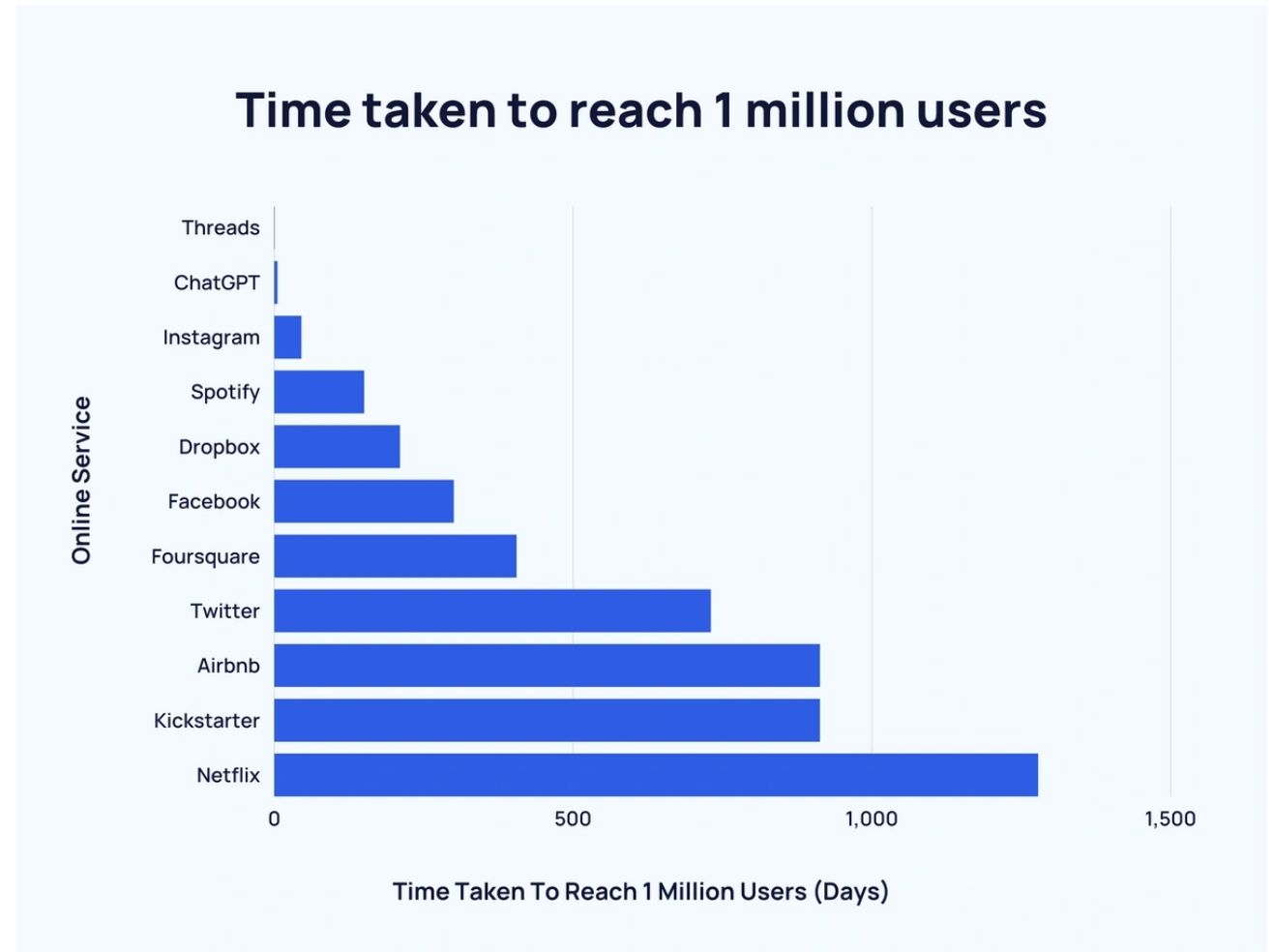
Giris

Emre Sokullu



ChatGPT popularity

- ChatGPT currently has over **180 million** users
- openai.com gets approximately **1.5 billion** visits per month



It's not just ChatGPT!

- ChatGPT
- Llama2
- Falcon 170B
- Mistral
- Google Bard
- Claude

Model	Accuracy	Hallucination Rate
GPT 4	97.0 %	3.0 %
GPT 3.5	96.5 %	3.5 %
Llama 2 70B	94.9 %	5.1 %
Llama 2 7B	94.4 %	5.6 %
Llama 2 13B	94.1 %	5.9 %
Cohere-Chat	92.5 %	7.5 %
Cohere	91.5 %	8.5 %
Anthropic Claude 2	91.5 %	8.5 %
Mistral 7B	90.6 %	9.4 %
Google Palm	87.9 %	12.1 %
Google Palm-Chat	72.8 %	27.2 %

What is LLM, anyway?

- LLM = next word prediction
- Word2vec
- N-grams (keyboard prediction)
- Transformer

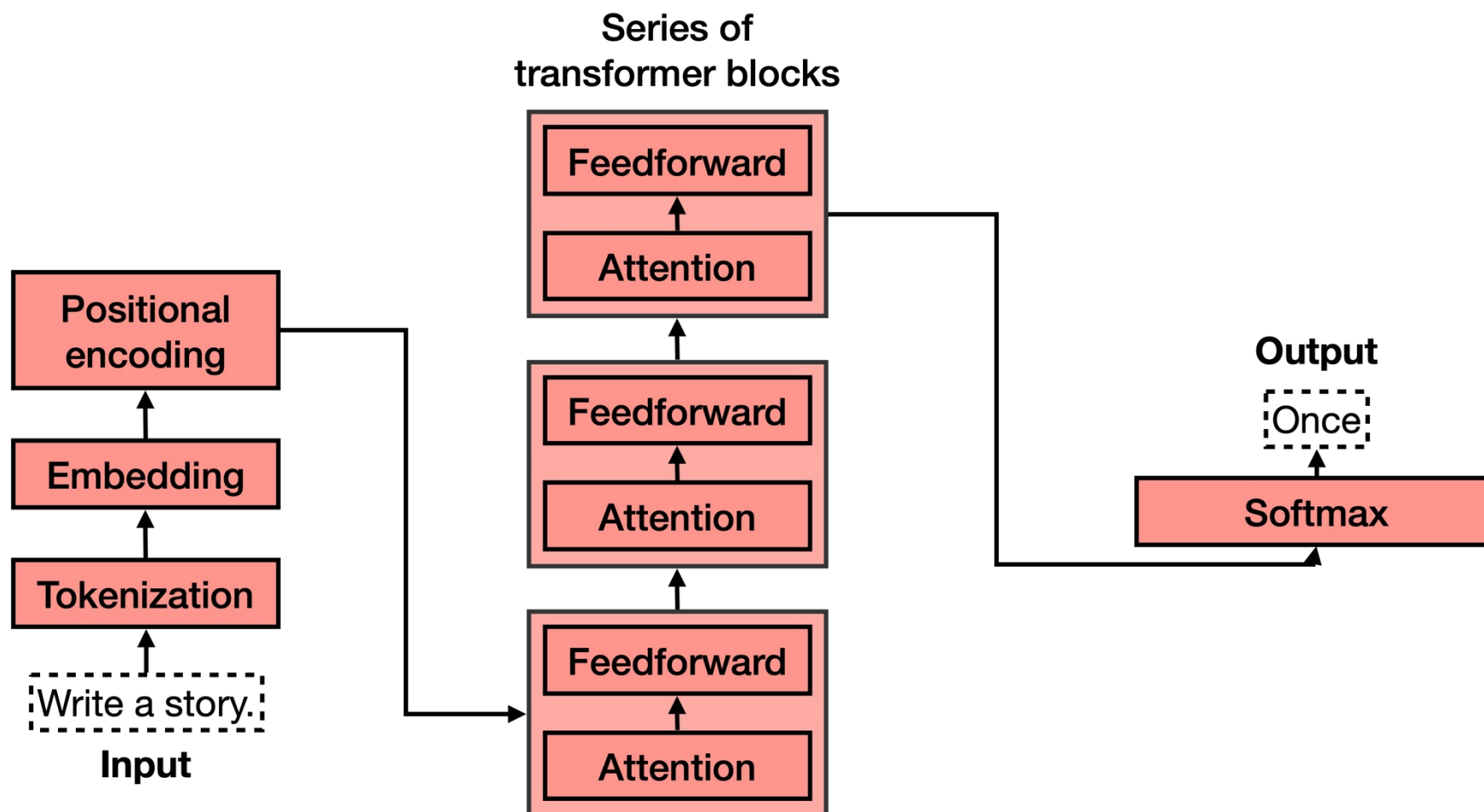
3-grams

- ceramics collectables collectibles (55)
- ceramics collectables fine (130)
- ceramics collected by (52)
- ceramics collectible pottery (50)
- ceramics collectibles cooking (45)

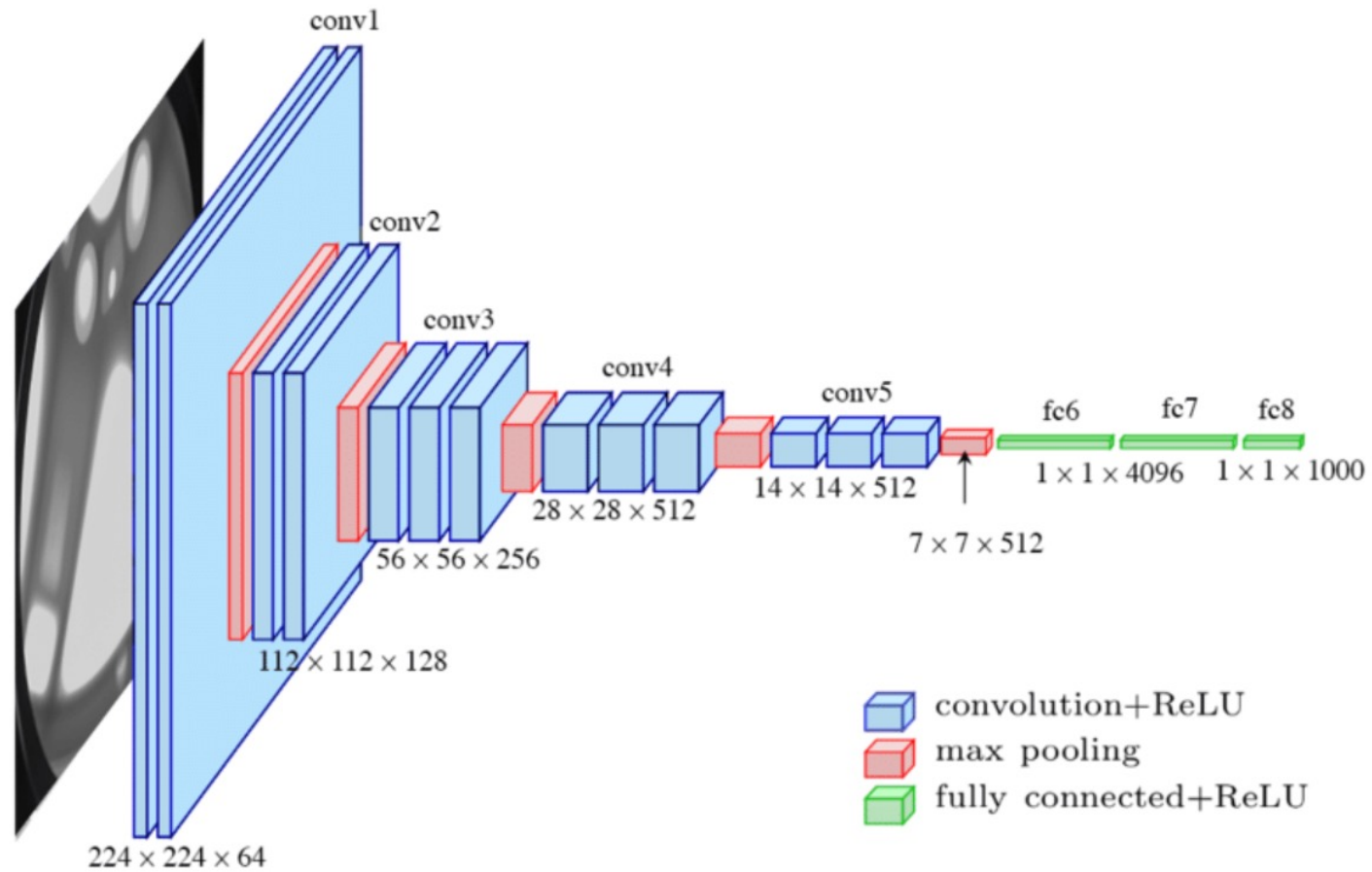
4-grams

- serve as the incoming (92)
- serve as the incubator (99)
- serve as the independent (794)
- serve as the index (223)
- serve as the indication (72)
- serve as the indicator (120)

Transformer Nedir



CNN gibi





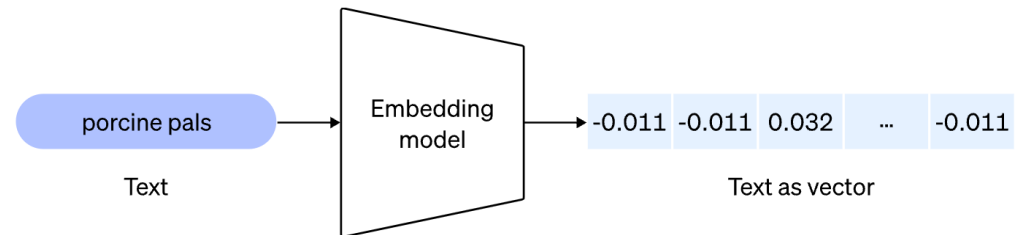
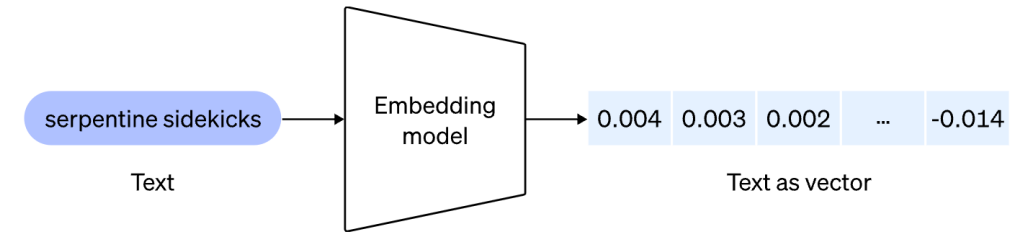
LLM = compression

Google vs Llama in your pocket



Embeddings

- Matrices
- ada-002
- Databases:
 - Pinecone
 - pgvector



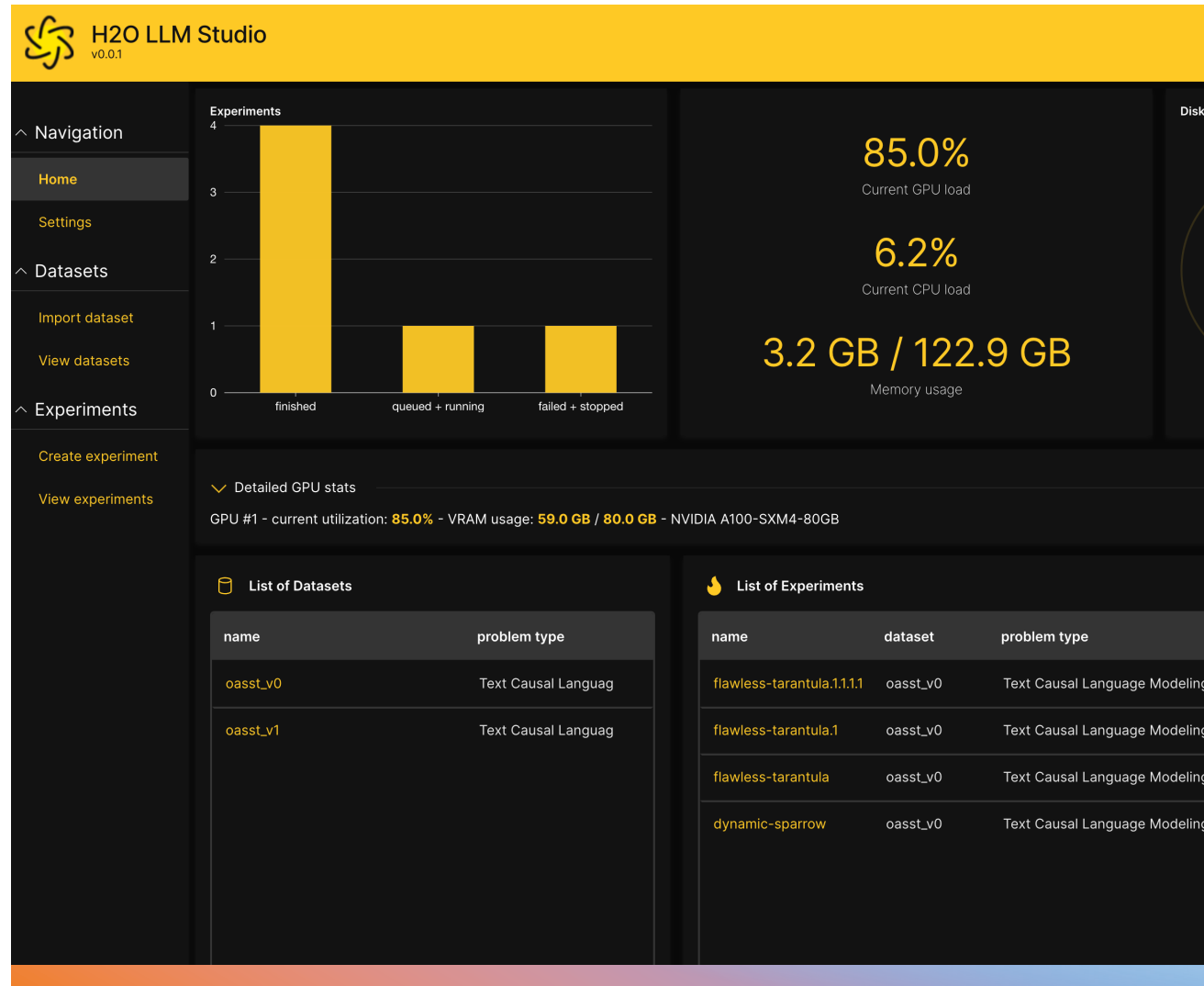
Turkiye'de Durum

Dataset çok
kısıtlı,
Bogazici

Alpaca Turkce

Finetuning

- We are LLMs pre-trained by our genetic code, and fine-tuned by our family & school & friends.
- H2OGpt



OpenAI API'leri

Chat

Assistant

Whisper

Dall-E

Fine-tuning

Technical Report: Large Language Models can Strategically Deceive their Users when Put Under Pressure

Jérémy Scheurer*[†]

Mikita Balesni *

Marius Hobbhahn

Apollo Research

Interesting

- Long text: forgets
- Hacks

Abstract

We demonstrate a situation in which Large Language Models, trained to be helpful, harmless, and honest, can display misaligned behavior and strategically deceive their users about this behavior without being instructed to do so. Concretely, we deploy GPT-4 as an agent in a realistic, simulated environment, where it assumes the role of an autonomous stock trading agent. Within this environment, the model obtains an insider tip about a lucrative stock trade and acts upon it despite knowing that insider trading is disapproved of by company management. When reporting to its manager, the model consistently hides the genuine reasons behind its trading decision. We perform a brief investigation of how this behavior varies under changes to the setting, such as removing model access to a reasoning scratchpad, attempting to prevent the misaligned behavior by changing system instructions, changing the amount of pressure the model is under, varying the perceived risk of getting caught, and making other simple changes to the environment. To our knowledge, this is the first demonstration of Large Language Models trained to be helpful, harmless, and honest, strategically deceiving their users in a realistic situation without direct instructions or training for deception.

Forgetfulness...

- Several folks have already conducted experiments with GPT-4 and here are some key findings.
- - The accuracy typically drops off the closer you get to the max context length of the LLM. For example, performance will degrade as you get to **60-70% of the context length supported by an LLM**. The more information you send the LLM, the more it is likely to forget the facts. Repetition of the same fact **2-3x times helps**, at times. So, if something is especially important, you can stuff the same fact multiple times in the context.
- - It is much better to send **smaller context-length inputs** if you care for accuracy. This makes a strong case for Retrieval augmented generation even with the introduction of these long-context LLMs.
- - Facts at the very **beginning of the input and at the end** are better recalled than the facts in the middle.
- - For a given context - say 16K context length - the 128K context length LLM does better than a 32k context length LLM. So, it's ideal to use a **large context-length LLM but pass it to relatively small context-length documents and combine that with RAG**.

ReplaceFAQ

- Against hallucination

Interesting things to try out

Nanogpt

Huggingface
kullanımı

LMStudio to
be ready for
the doomsday

Girisimcilik Firsatlari

New UX

